

Ethics of Generative & Virtual Experiences in Medicine

JUNAID QADIR

Professor of Computer Engineering
Department of Computer Science and Engineering
Qatar University, Doha, Qatar

Learning Objectives & Conflict of Interest Statement

I do not have an affiliation (financial or otherwise) with any for-profit or not-for-profit organizations.
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At the end of this session, participants will be able to:

1. Identify the key components of AI-enabled metaverses, specifically focusing on the integration of GenAI and XR in healthcare.
2. Describe the potential opportunities and inherent risks associated with the use of AI and XR in medical practice and patient care.
3. Apply ethical frameworks to assess and guide the implementation of healthcare metaverses in clinical settings.

Outline

1. Introduction to AI-Enabled Metaverses (using GenAI and XR)
2. Opportunities and Risks
3. Ethics of Healthcare Metaverses

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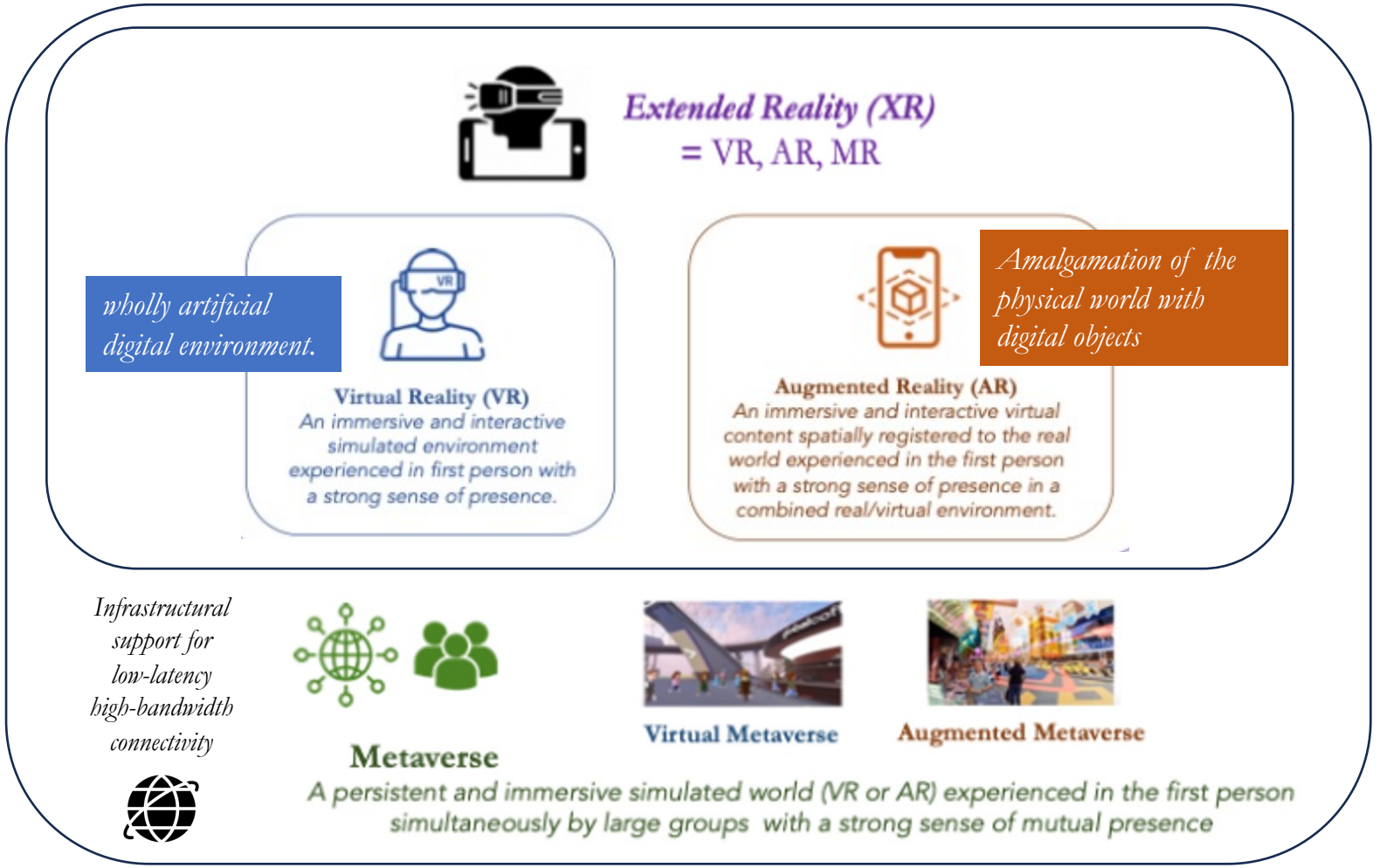
Introduction to GenAI, AR, VR, and the Metaverse



The metaverse is the 3D version of the global internet we know, powered by AI and primarily accessed using XR devices. In a formula:

$$\text{Metaverse} = \text{Internet}(3D)^{\text{AI}} \leftrightarrow \text{XR}$$

Introduction to XR



Enabling Technologies:

- *Spatial Computing*
- *Spatial Web*
- *Haptic Technology*
- *Localization and Mapping*
- *AR/VR Libraries*
- *Gaming Industry*

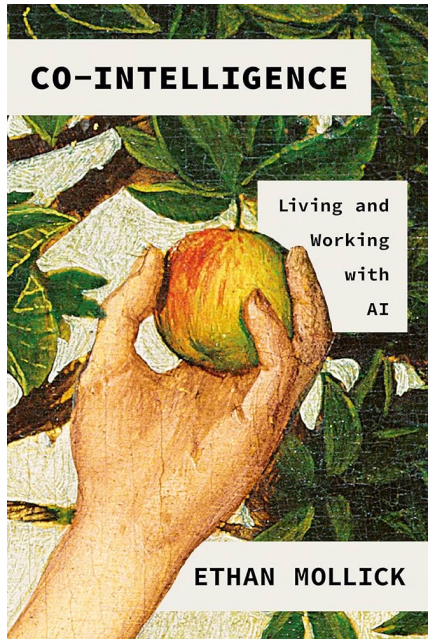
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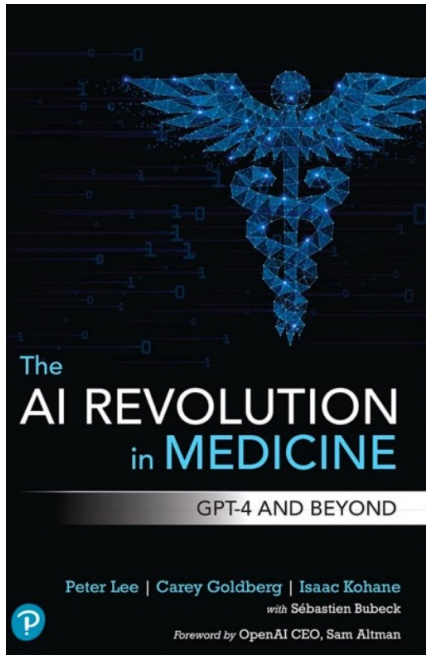
Vision Pro

Introduction to AI/ GenAI

MOTHERBOARD
TECH BY VICE

ChatGPT Is Passing the Tests Required for Medical Licenses and Business Degrees





Roles AI/ GenAI can play in metaverses:

—*Conversational interface*
—*Natural language processing (NLP)*
—*Computer vision (emotion recognition)*
—*Digital twins (virtual data generation)*
—*Understanding user's activities*
—*Avatar generation, etc.*

Open-Source Technology & the Openness Dilemma

"Open-Source" AI Models

MISTRAL AI_

Llama 3

Claude

LLAMA 2

VS.

Proprietary Models

OpenAI ChatGPT

There are risks in opening up these highly capable "foundation models";
But lack of transparency and openness is also problematic.

Is GenAI reliable and trustworthy?

(the risk of working with blackbox models)





Sam Altman

@sama

ChatGPT is incredibly limited, but good enough at some things to create a misleading impression of greatness.

it's a mistake to be relying on it for anything important right now. it's a preview of progress; we have lots of work to do on robustness and truthfulness.

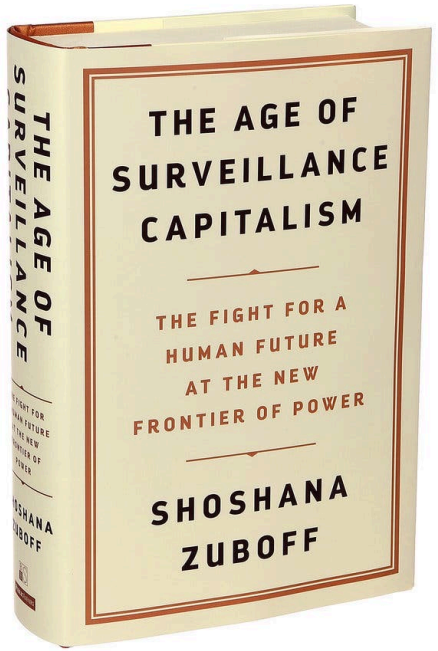
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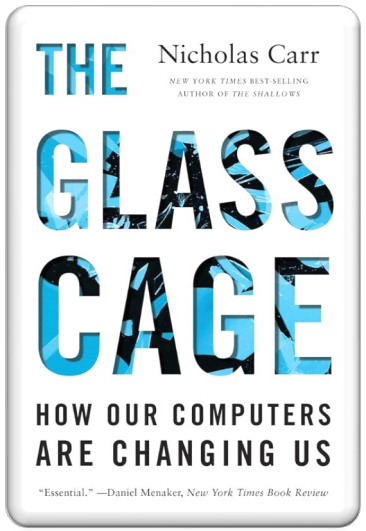
CAVEAT EMPTOR:

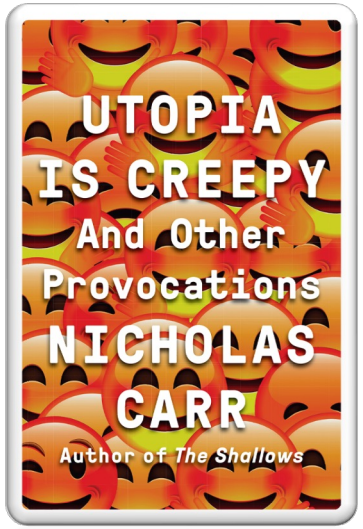
Let's say it's highly accurate (e.g., 90% of the time, it is factual). Can we use it for critical tasks?

***The problem is that we do not know which 90%**, and worse, ChatGPT doesn't know either!"*

Metaverse and our relationship with our self/ world







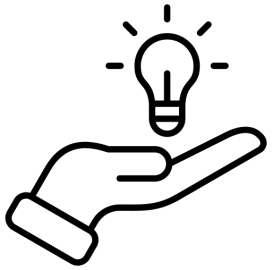
Technology as a **tool**	*VS.*	Technology as a **media**
Tool users		*Attention as a commodity to be captured and sold*

The cost we pay for free services
(e.g., privacy & focus)

Does the technology strengthen or weaken our relationship with ourselves, others, and the world?

2

Opportunities and Risks



Opportunities

Immersive Education

Unique first-person immersive experience to enhance human perception and educate in the relevant environment.

Emotional Support

AR/VR can be used for emotional support, and for managing stress and anxiety

Accessibility and Inclusiveness

Customization for visually-impaired people with enhanced emphasis on other senses with support from haptic or audio interface.

Empathy and Awareness

Increase empathy and promote awareness of problems such as racism, war, and discrimination

Simulations and What-If-Analysis

Simulation and counterfactual analysis to calibrate diagnosis & plan treatments

Risks



Deskilling

Potential loss of professional skills due to over-reliance on AI and automation.

***Example**: Healthcare professionals relying too much on AI diagnostics, reducing their hands-on experience and decision-making skills.*



Disrupting Human Interactions

AI and virtual environments may reduce face-to-face interactions, impacting care.

Example: Virtual consultations replacing in-person visits, leading to a lack of personal connection between patients and healthcare providers.



Privacy and Surveillance

Increased data collection/ monitoring could lead to privacy and ethical concerns.

Example: Continuous patient tracking through wearable devices & virtual platforms.



Mental Health Concerns

Extended use of virtual environments could impact mental well-being.

Example: Patients experiencing isolation or addiction to virtual healthcare platforms.

Risks



Digital Divide and Equitable Access

Disparities in access to technology could exacerbate inequalities in healthcare.

Example: Underprivileged populations lacking access to the necessary technology to benefit from metaverse healthcare services.



Concentration of Power

Dominance by a few large corporations could limit competition and innovation.

Example: Major tech companies controlling the healthcare metaverse, reducing diversity in service providers.



Manipulative Practices

AI/ virtual environments could be used to manipulate user behavior and decisions.

Example: Algorithms designed to promote certain treatments or products for profit rather than patient benefit.



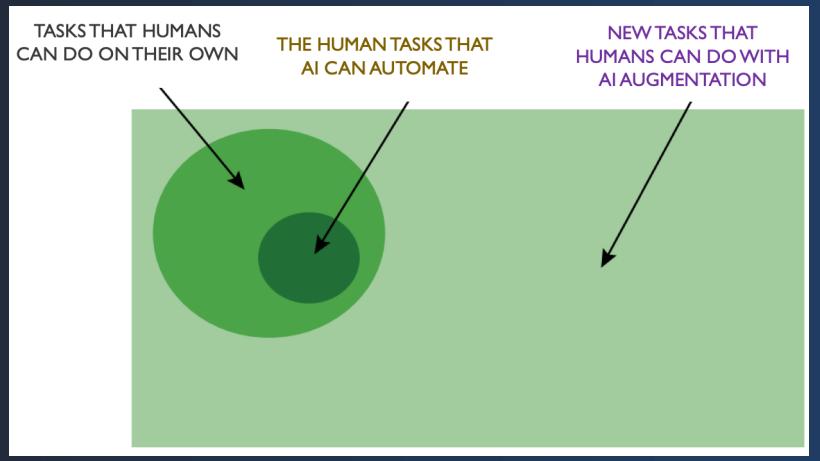
Hyperreality and Ethical Concerns

Difficulty distinguishing between virtual and real experiences may raise ethical issues.

Example: Patients receiving treatments in virtual/generative environments that they perceive as real, leading to confusion or mistrust.

3

Ethics of Generative/Virtual Experiences



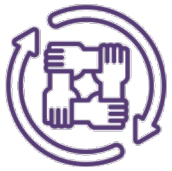
AI-XR Healthcare Metaverses should be: (1) Human-Centered; and (2) Agency, Relationship, and Capability Enhancing (or at least conserving).

Guidelines for Judging AI-XR Metaverse



Prioritize Education Reform and Workforce Development

Redesign training programs to emphasize human strengths that complement AI, minimizing workforce displacement.



Identify Synergies Rather Than Replacement

Focus on applications where AI & human expertise complement, addressing clinical expertise shortages.



Engage Human Abilities Through AI Systems

Automate mundane tasks with AI, allowing clinicians to focus on relationship-building, counseling, and management.



Expand Reach Through Automated Systems

Use AI/ telemedicine to reach patients outside traditional settings, providing management & care for chronic conditions.

Recommended AI-XR Best Practices



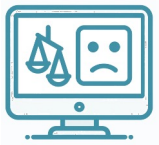
Beware of Marketing Hype: Recognize Real Opportunities

Avoid overhyping AI in healthcare metaverses. Focus on addressing care variability, high costs, and limited access.



Demand Transparency in Data Collection and Algorithm Evaluation

Maintain transparency in data collection & evaluation. Balance innovation with safety & regulation.



Identify and Correct Biases in Decision-Making Tools

Allocate resources to identify and mitigate biases in AI tools. Ensure algorithms are fair and just.



Robust Evaluations of Model Performance

Thoroughly evaluate models for performance, utility, vulnerabilities, and bias across different populations.



Develop AI Systems with Security in Mind

Design AI systems considering adversarial threats, inspired by cybersecurity defenses.



Some Questions to Consider In Healthcare Metaverses

- **Does XR/AI *enhance human engagement* with the physical world and real-world healthcare issues?**
- ***Will the XR/AI amplify existing real-world inequality in healthcare access and outcomes?***
- **Have the *system-level repercussions of XR/ AI technology* usage in medical practice been considered?**
- **Which policy and regulatory frameworks are in place to ensure human rights and autonomy when XR/AI is used in healthcare metaverses?**
- **Have the necessary steps been taken to accommodate and promote *diversity and inclusion* (D&I) so that the developed technology is appropriate for the needs and sensitivities of different patient groups?**
- **Is there transparency in the system regarding what data is collected, who it is shared with and how it is used in medical contexts?**
- **Has appropriate consideration been given to *safeguarding children's rights* if the application is intended for use in by pediatric healthcare?**
- **Is the technology culturally appropriate for diverse patient populations in healthcare settings?**

Thank you!

Your comments and input are most welcome

jqadir@qu.edu.qa